

Weekly Report

Physics, controllers and benchmark

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2026-05-03

What was accomplished this week

1. **Physics of the 2D simulator.**
2. **3 Drone Swarm controllers implemented** — Potential Fields, Consensus-Based Coordination, and Multi-Agent RL.
3. **Comparison study.** 144 runs across 3 hand-drawn maps × 4 swarm sizes × 4 policies × 3 seeds

Questions / Challenges

The most difficult and time-consuming part was ensuring the physics engine is correctly modeled. I would say this is the most important piece also, since if the physics is not correct, no optimization algorithm would work properly on it.

Goals for next week

Do a full e2e run with what Raffi and Elen did, to understand which algorithm will perform better for the given case.

1. Physics fix - what our drones have

Each drone is a **Hawk's Work F450**

Component	Spec	Mass
Frame	F450, 450 mm wheelbase	280 g
Motors (4x)	A2212 920 KV brushless	208 g
ESCs (4x)	20 A brushless	71.6 g
Propellers (4x)	9450 self-tightening	40 g
Battery	11.1 V 3S LiPo, 4200 mAh, 25 C	330 g
Flight controller	Pixhawk 2.4.8 (PX4 autopilot)	15.8 g
Companion computer	NVIDIA Jetson Nano	178 g
Camera	e-con STEEReoCAM Nano	159 g
Total		≈ 1.3 kg

2. How did I do the tests?

2.1. Test maps

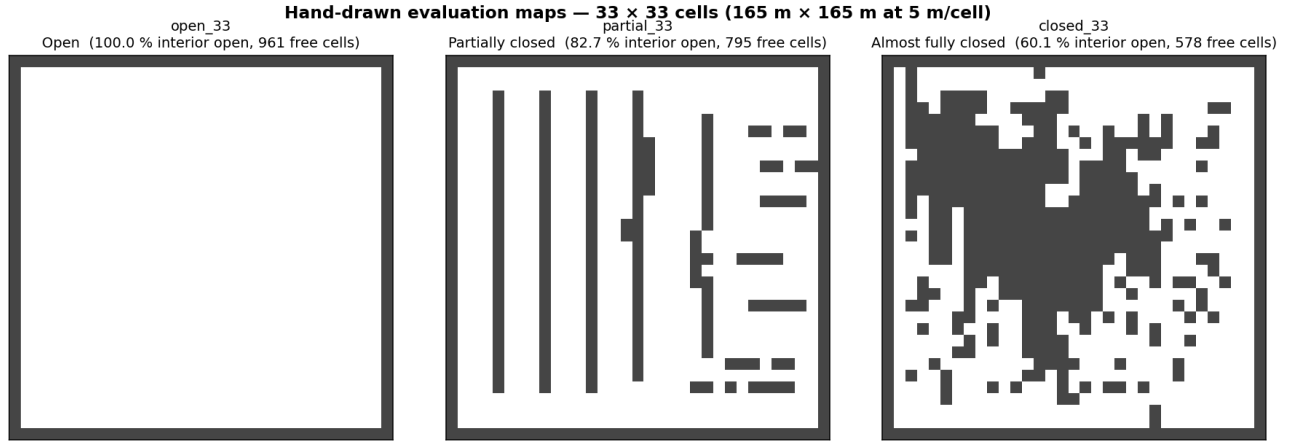


Figure 1: Hand-drawn evaluation maps

2.2. Swarm sizes

Tested four drone counts: **n_drones in {2, 5, 10, 20}**. Each drone is the same Hawk's Work F450 build (1.3 kg, 11.1 V 3S 4200 mAh LiPo, STEEReoCAM Nano forward-facing wedge sensor).

2.3. Algorithms implemented

Controller	Type	State	Training
Random	baseline	stateless	none
PF	classical (potential fields)	stateless	none
Consensus	classical (decentralized)	stateless	none
MARL	reinforcement learning	stateful (NN weights)	150K PPO steps × 4 parallel envs

Potential Fields. Khatib-style superposition of three forces:

$$F_{\text{total}} = \text{attract_to_nearest_uncovered} + \text{repulse_from_drones} + \text{repulse_from_walls}$$

Consensus.

Each drone polls neighbors; the communication group implicitly partitions uncovered cells by Euclidean distance, and each drone heads to the **nearest uncovered cell in its own Voronoi region**. Repulsion identical to PF.

MARL.

Independent PPO with shared parameters via Stable-Baselines3. The joint policy network sees per-drone local state (own pos/vel/heading, battery fraction, 5×5 local coverage mask, global coverage fraction) flattened across the swarm; emits joint 2D acceleration.

2.4. Benchmarks

3 maps × 4 drone counts × 4 policies × 3 seeds = **144 runs**

Final coverage (mean of 3 seeds)

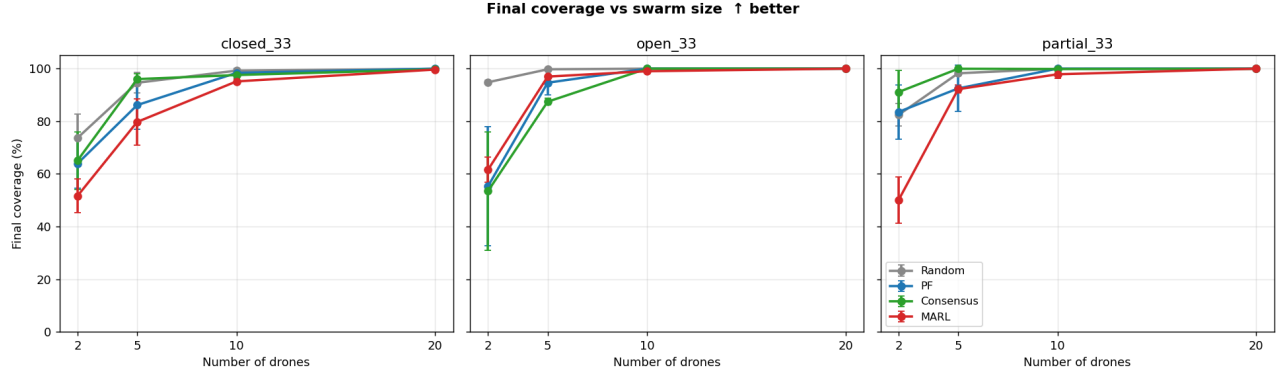


Figure 2: Final coverage vs swarm size

From here we can already observe that **n = 10 is the sweet spot**. For a 165 × 165 m world (24,025 m² navigable interior in the open case), 10 drones is enough.

open_33 — open arena

n_drones	Random	PF	Consensus	MARL
2	94.8 %	55.4 %	53.5 %	61.6 %
5	99.8 %	94.7 %	87.4 %	97.0 %
10	100.0 %	100.0 %	100.0 %	99.0 %
20	100.0 %	100.0 %	100.0 %	100.0 %

partial_33 — scattered obstacles

n_drones	Random	PF	Consensus	MARL
2	82.4 %	83.6 %	91.1 %	50.1 %
5	98.2 %	92.5 %	100.0 %	92.2 %
10	100.0 %	100.0 %	99.9 %	97.9 %
20	100.0 %	100.0 %	100.0 %	100.0 %

closed_33 — maze-like

n_drones	Random	PF	Consensus	MARL
2	73.7 %	64.0 %	65.1 %	51.7 %

n_drones	Random	PF	Consensus	MARL
5	94.6 %	86.2 %	96.0 %	79.8 %
10	99.3 %	98.4 %	97.6 %	95.2 %
20	99.9 %	100.0 %	99.7 %	99.7 %

Time to 80 % coverage

Consensus is the fastest to 80 % at $n \geq 5$ across all three maps. PF catches up at $n = 20$. MARL is consistently the slowest because its trained policy emits less aggressive forward motion than the analytic potential gradient.

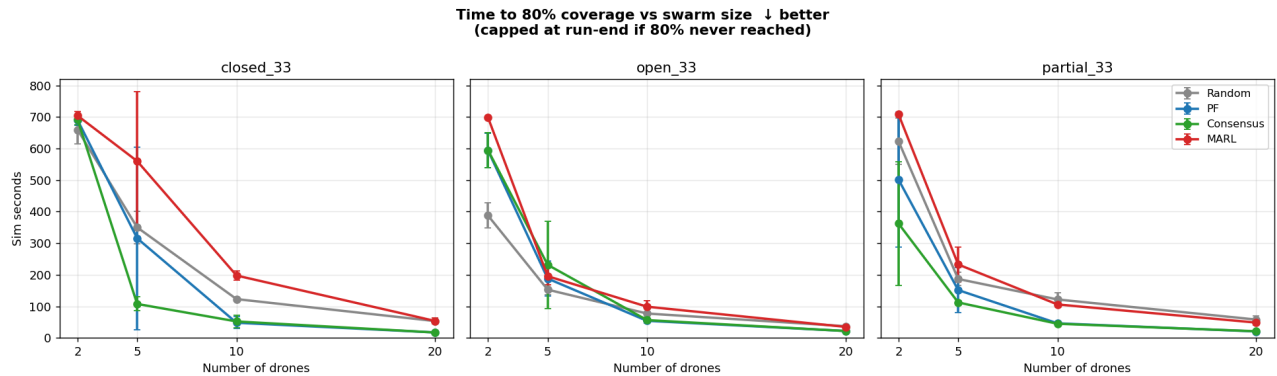


Figure 3: Time to 80 % vs swarm size

Re-coverage waste (wasted entries per % coverage)

PF wins on efficiency, especially at large swarms. This is because in PF each drone heads to its nearest *uncovered* cell, so once a cell is covered the target moves forward and the drone rarely backtracks.

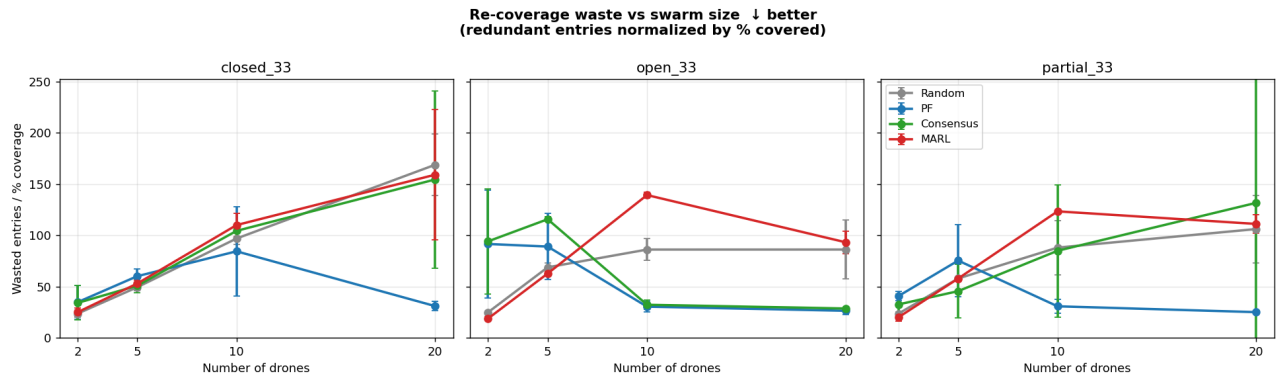


Figure 4: Re-coverage waste

Coverage curves over time — full grid

MARL — a reinforcement learning method — never wins. I was surprised by this. It could be either the training time I gave, or the parameters. But overall, I expected better results from it.

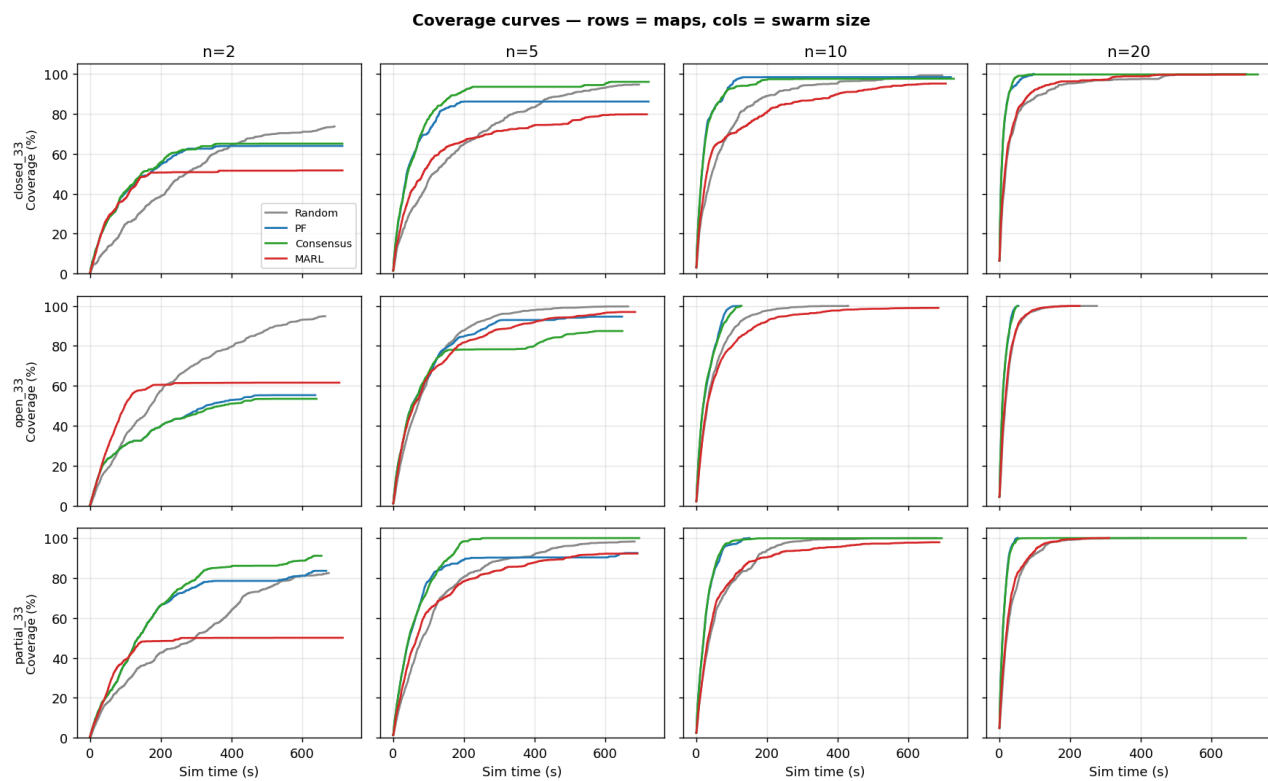


Figure 5: Coverage curves grid